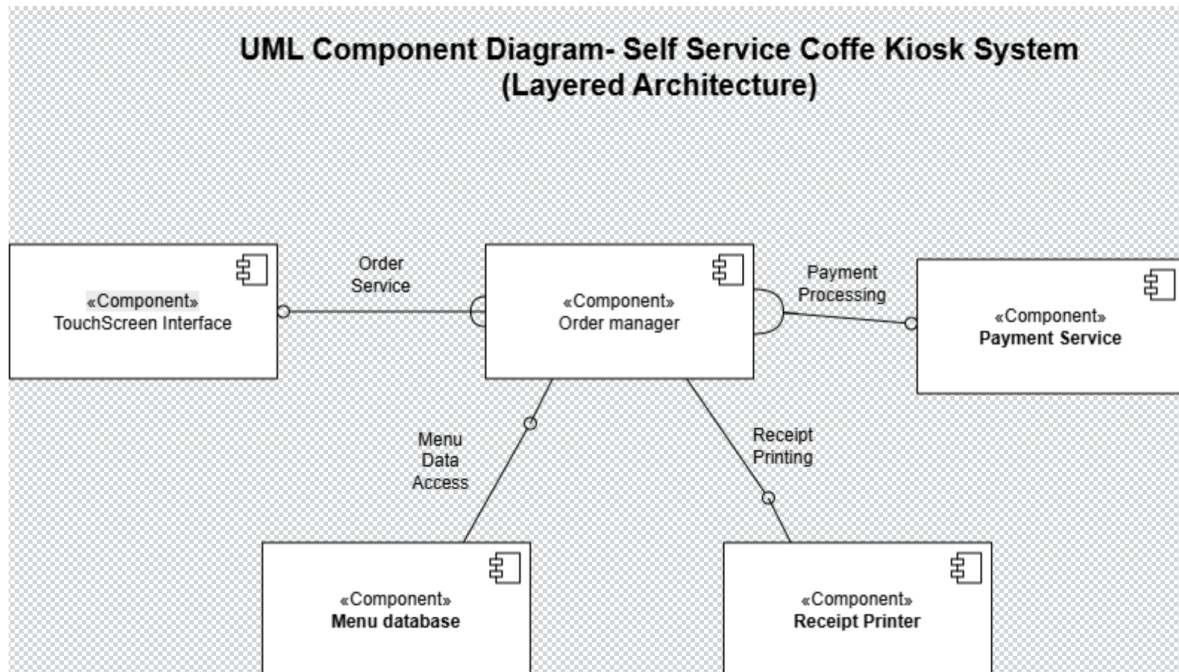


UML METHOD:



Architecture Justification – Self-Service Coffee Kiosk System

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

The system is divided into separate layers/components such as the **Touchscreen Interface, Order Manager, Payment Service, Menu Database, and Receipt Printer**. This separation makes the system easier to understand, develop, and maintain.

Reason 1 – Separation of Responsibilities:

The Touchscreen Interface handles user interaction, while the Order Manager handles order processing. The Payment Service manages credit-card payments, and the Menu Database stores menu and pricing information. This separation keeps each component focused on its specific responsibility.

Reason 2 – Hardware and Data Management:

The architecture allows the kiosk to communicate with the required hardware, such as the touchscreen and receipt printer, while keeping menu and pricing information separately managed in the database. This makes the system easier to modify when hardware or menu information changes.

Security Advantage:

Layered separation helps isolate sensitive payment processing from the user-interface layer. The Payment Service can handle credit-card transactions without exposing payment-related processing directly to the touchscreen component.

Performance Benefit:

Separating responsibilities allows each component to perform its specific task efficiently. Menu and

pricing information can be retrieved directly from the Menu Database, while order processing and payment processing are handled by their respective components, reducing unnecessary processing in the interface layer.

Conclusion:

Therefore, Layered Architecture is suitable for the Self-Service Coffee Kiosk System because it provides clear separation of responsibilities, supports the required hardware and data management, and offers benefits for security, maintainability, and performance.